

1 Spin and the Nuclear Magnetic Resonance Phenomenon 自旋與核磁共振現象

Medical magnetic resonance (MR) imaging uses the signal from the nuclei of hydrogen atoms (1H) for image generation.

醫學磁共振（MR）成像利用氫原子核（ 1H ）的訊號來生成影像。

A hydrogen atom consists of a nucleus containing a single proton and of a single electron orbiting the nucleus (► Fig. 1). The proton having a positive charge and the electron a negative charge, the hydrogen atom as a whole is electrically neutral. The proton is of interest here.

氫原子由包含單一質子的原子核，以及繞原子核運行的單一電子所組成（► 圖 1）。質子帶有正電荷，而電子帶有負電荷，因此氫原子整體呈電中性。此處所關注的是質子。

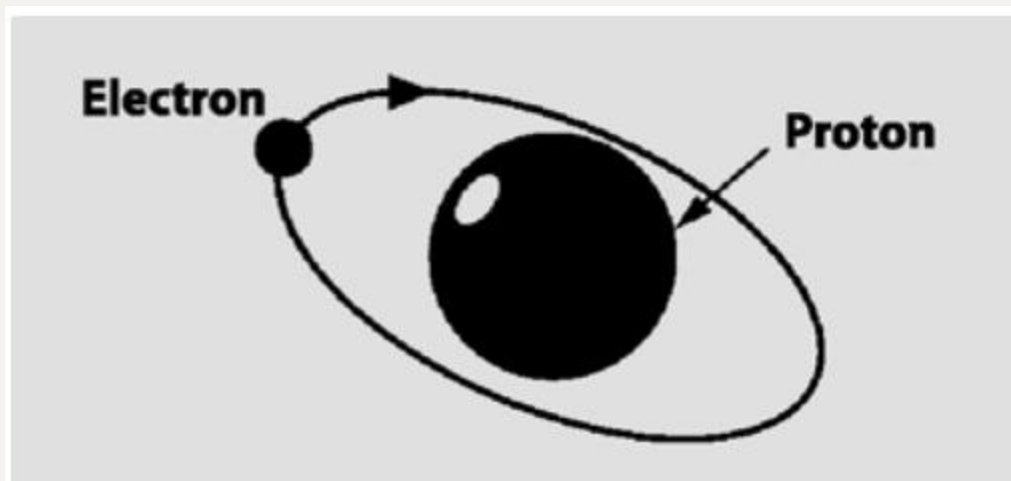


Fig. 1.

圖 1。

Apart from its positive charge, the proton possesses spin, an intrinsic property of nearly all elementary particles. This means that the proton rotates about its axis like a spinning top. Such a proton has two important properties:

除了帶有正電荷之外，質子還具有自旋（spin），這是幾乎所有基本粒子都具備的內在性質。這表示質子會像陀螺一樣繞著自身的軸旋轉。這樣的質子具有兩項重要性質：

As a rotating mass (m), the proton has angular momentum and acts like a spinning top that strives to retain the spatial orientation of its rotation axis (► Fig. 2a).

作為旋轉質量（ m ），質子具有角動量，其作用就像一個陀螺，力求保持其旋轉軸的空間取向（► 圖 2a）。

As a rotating mass with an electrical charge, the proton additionally has magnetic moment (B) and behaves like a small magnet. Therefore, the pro-

作為帶有電荷的旋轉質量，質子還具有磁矩（ B ），並且表現得像一個小磁鐵。因此，質-

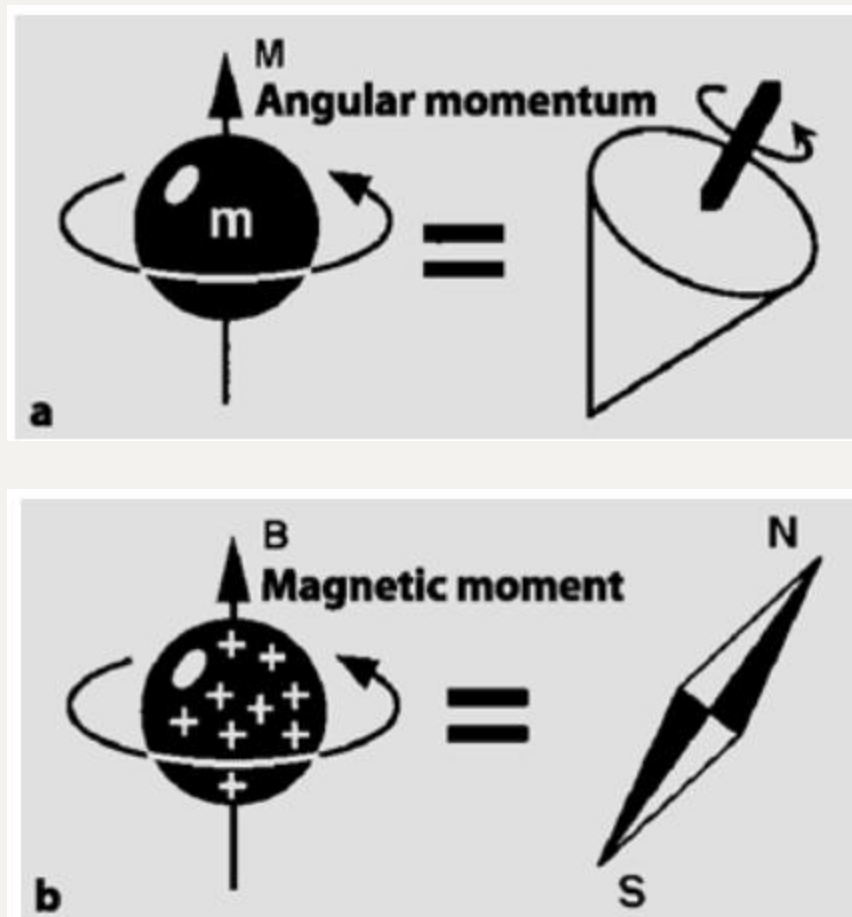


Fig. 2.

圖 2。

ton is affected by external magnetic fields and electromagnetic waves and, when it moves, induces a voltage in a receiver coil (► Fig. 2b).

ton 會受到外部磁場與電磁波的影響，並在移動時於接收線圈中感應出電壓（► 圖 2b）。

A hydrogen nucleus differs from a spinning top, however, in that we cannot look into it and thus cannot see its intrinsic angular momentum, or spin, from the outside. In this respect, the nucleus is a black box for us. Nevertheless, we can identify the orientation of its rotation axis from the magnetization vector B . Thus, when we describe the rotation of a proton, we are not referring to its (invisible) angular momentum but to the “visible” motion of its magnetic axis, B . This motion is visible, so to speak, because it generates a signal in a receiver coil just like a magnet does in an electrical generator (e.g. a bicycle dynamo).

然而，氫原子核與陀螺不同之處在於，我們無法觀察其內部，因此也無法從外部看見其內稟角動量，亦即自旋。在這方面，原子核對我們而言就像一個黑盒子。儘管如此，我們仍能透過磁化向量 B 確定其旋轉軸的方向。因此，當我們描述質子的旋轉時，所指的並不是其（不可見的）角動量，而是其磁軸 B 的「可見」運動。之所以說這種運動是可見的，是因為它會在接收線圈中產生訊號，就如同磁鐵在發電機（例如自行車發電機）中產生訊號一樣。

There is another, very important difference: while a spinning top can be slowed down and thus finally comes to a standstill, a proton's spin always has the same magnitude and can neither be accelerated nor decelerated, precisely because it is a fundamental property of elementary particles. Spin is simply there all the time!

還有另一個非常重要的差異：陀螺旋轉時可以逐漸減速，最終停止不動；然而，質子的自旋始終具有相同的大小，既不能加速，也不能減速，正是因為自旋是基本粒子的一項基本性質。自旋始終存在！

How will a spin behave when brought into a strong magnetic field? To answer this question, let us again consider the spinning top:

當自旋置於強磁場中時會如何運動？為了回答這個問題，讓我們再次考慮陀螺：

When an external force (typically the earth's gravitational field G) acts on a spinning top and tries to alter the orientation of its rotational axis, the top begins to wobble, a process called precession. At the same time, friction at the point of contact withdraws energy from the spinning top and slows down its rotation. As a result, its axis becomes more and more inclined and the top finally falls over (► Fig. 3).

當外力（通常是地球的重力場 G ）作用於旋轉陀螺，並試圖改變其旋轉軸的方向時，陀螺便會開始搖擺，這個過程稱為進動（precession）。與此同時，接觸點的摩擦力會從旋轉中的陀螺帶走能量，使其旋轉速度減慢。因此，其旋轉軸會變得越來越傾斜，陀螺最終會翻倒（► 圖 3）。

Once again, back to our hydrogen nuclei: when these are exposed to an external magnetic field, B_0 , the magnetic moments, or spins, align with the direction of the field like compass needles. The magnetic moments do not only align with the field but, like spinning tops, undergo precession (► Fig. 4). Precession of the nuclei occurs at a characteristic speed that

再次回到氫原子核：當這些原子核暴露於外部磁場 B_0 時，磁矩或自旋會像指南針的指針一樣，沿著磁場方向排列。磁矩不僅會與磁場對齊，還會像旋轉的陀螺一樣進行進動（► 圖 4）。原子核的進動會以一種特徵速度發生，這種速度 天天中彩票nba

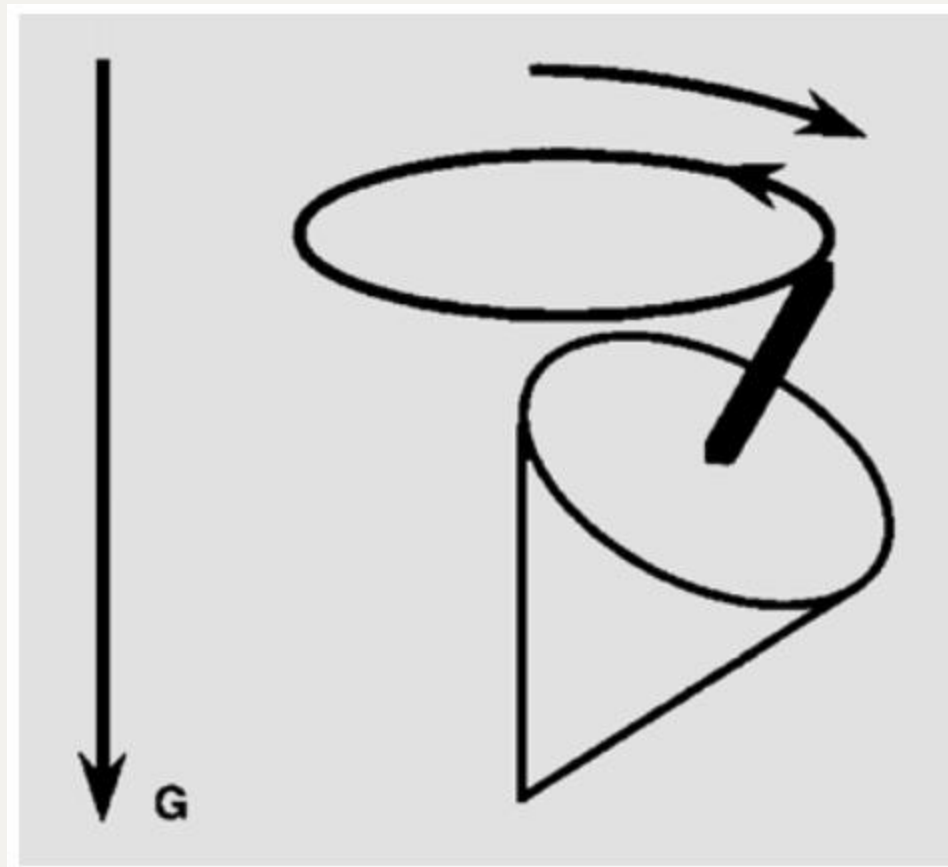


Fig. 3.

圖 3。

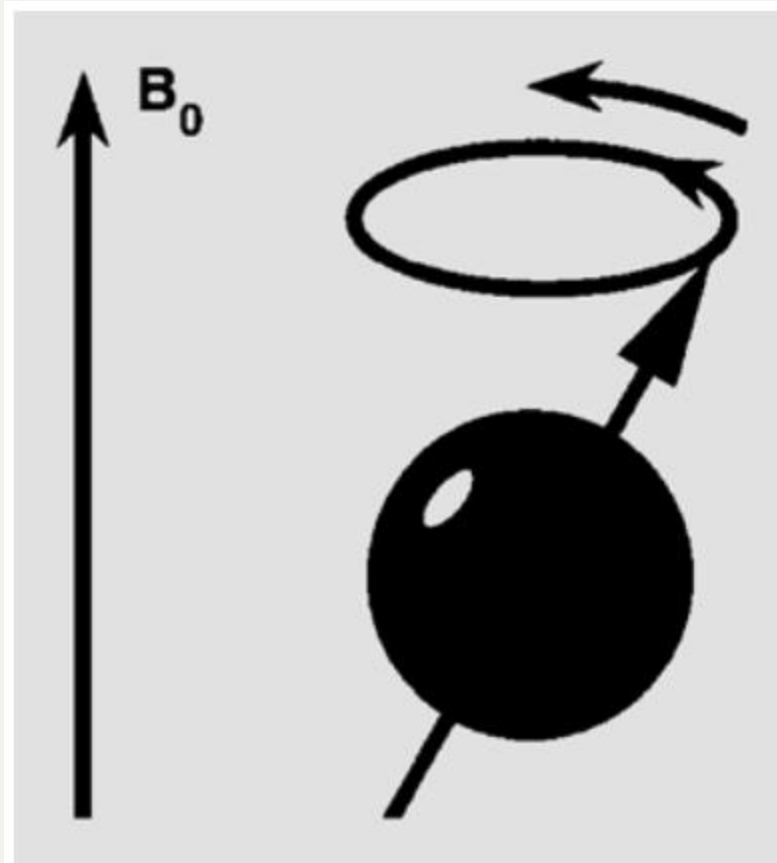


Fig. 4.

圖 4。

is proportional to the strength of the applied magnetic field and is called Larmor frequency. Alignment of the spins parallel to the magnetic field is a gradual process and, as with spinning tops, is associated with the dissipation of energy (► Chapter 2.1).

與所施加磁場的強度成正比，稱為拉莫爾頻率。自旋平行於磁場的排列是一個逐漸進行的過程，且如同陀螺旋轉一般，伴隨著能量的耗散（► 第 2.1 章）。

The Larmor frequency is a very important concept that is at the core of MR imaging. Let us therefore repeat:

拉莫爾頻率是磁共振成像（MR imaging）的核心概念，非常重要。因此，讓我們再次重申：

The Larmor or precession frequency is the rate at which spins wobble when placed in a magnetic field.

Larmor 或進動頻率是自旋置於磁場中時擺動的速率。

The Larmor frequency is directly proportional to the strength (B_0) of the magnetic field and is given by the Larmor equation:

Larmor 頻率與磁場強度 (B_0) 成正比，並由 Larmor 方程式給出：

$$\omega_0 = \gamma_0 \cdot B_0 \text{ where}$$

$$\omega_0 = \gamma_0 \cdot B_0, \text{ 其中}$$

- ω_0 is the Larmor frequency in megahertz [MHz],
- γ_0 the gyromagnetic ratio, a constant specific to a particular nucleus, and
- B_0 the strength of the magnetic field in tesla [T].

- ω_0 是以兆赫茲 [MHz] 為單位的拉莫爾頻率，
- γ_0 是旋磁比，為特定原子核所特有的常數，而
- B_0 是以特斯拉 [T] 為單位的磁場強度。

Protons have a gyromagnetic ratio of $\gamma = 42.58 \text{ MHz/T}$, resulting in a Larmor frequency of 63.9 MHz at 1.5 T as opposed to only about 1 kHz in

質子的磁旋比為 $\gamma = 42.58 \text{ MHz/T}$ ，因此在 1.5 T 下的拉莫爾頻率為 63.9 MHz，相較之下在其中僅約為 1 kHz。

the magnetic field of the earth (by comparison, FM radio transmitters operate at 88–108 MHz).

地球的磁場（相比之下，FM 廣播發射器的運作頻率為 88–108 MHz）。

What happens to the spins precessing and slowly aligning inside the magnetic field? Let us see ...

在磁場內進動並緩慢對齊的自旋會發生什麼事？讓我們看看.....

While the spin system relaxes and settles into a stable state, longitudinal magnetization M_z is building up in the z-direction because the magnetic vectors representing the individual magnetic moments add together. This also happens in the earth's magnetic field but the resulting longitudinal magnetization is only weak. The magnetic field B_0 of an MR imager is 60,000 times stronger and the resulting longitudinal magnetization is correspondingly larger. Because the MR signal is very weak, magnetization must be large enough to obtain a signal at all. Actually, things are even a bit more complicated: the spins tend to align parallel or anti-

parallel to the magnetic field with parallel alignment being slightly preferred because it is equivalent to spins residing in a more favorable energy state. Hence, under steady-state conditions, a slightly larger fraction aligns parallel to the main magnetic field. It is this small difference that actually produces the measurable net magnetization M_z and is represented by the net magnetization vector (NMV). Since the energy difference between the two orientations depends on the strength of the external magnetic field, M increases with the field strength.

當自旋系統逐漸鬆弛並穩定至穩定狀態時，由於代表個別磁矩的磁向量彼此相加，縱向磁化 M_z 會沿 z 方向逐漸建立。在地球磁場中也會發生同樣的情況，但所產生的縱向磁化仍然相當微弱。MR 成像儀的磁場 B_0 強大 60,000 倍，因此所產生的縱向磁化也相應地更大。由於 MR 訊號非常微弱，磁化強度必須足夠大，才能獲得任何訊號。

事實上，情況甚至更加複雜：自旋傾向於與磁場平行或反平行排列，而平行排列略占優勢，因為這相當於自旋處於較有利的能量狀態。因此，在穩態條件下，會有略大比例的自旋與主磁場平行排列。正是這個微小差異實際上產生了可測量的淨磁化 M_z ，並由淨磁化向量（NMV）表示。由於兩種方向之間的能量差異取決於外部磁場的強度，因此 M 會隨著磁場強度增加而增大。

Energy can be introduced into such a stable spin system by applying an electromagnetic wave of the same frequency as the Larmor frequency. This is called the resonance condition. The required electromagnetic wave is generated in a powerful radio transmitter and applied to the object to be imaged by means of an antenna coil. The process of energy absorption is known as excitation of the spin system and results in the longitudinal magnetization being more and more tipped away from the z -axis toward the transverse (xy -) plane perpendicular to the direction of the main magnetic field.

透過施加與拉莫爾頻率（Larmor frequency）相同頻率的電磁波，可以將能量引入此類穩定的自旋系統中。這稱為共振條件（resonance condition）。所需的電磁波由大功率無線電發射器產生，並藉由天線線圈施加至待成像的物體。能量吸收的過程稱為自旋系統的激發（excitation），其結果是縱向磁化逐漸偏離 z 軸，朝向垂直於主磁場方向的橫向（ xy -）平面傾斜。

All of the longitudinal magnetization is rotated into the transverse plane by a radiofrequency (RF) pulse that is strong enough and applied long enough to tip the magnetization by exactly 90° (90° RF pulse). The resulting magnetization is now denoted by M_{xy} rather than M_z because it now lies in the xy -plane. Whenever transverse magnetization is present, it rotates or precesses about the z -axis, which has the effect of an electrical generator and induces an alternating voltage of the same frequency as the Larmor frequency in a receiver coil: the MR signal. This signal is collected and processed with sensitive receivers and computers to generate the MR image. The process of excitation of a spin system is illustrated graphically in ► Fig. 5.

所有縱向磁化皆會受到一個足夠強且施加時間足夠長的射頻（RF）脈衝作用，使磁化方向恰好偏轉 90° （ 90° RF 脈衝）而旋轉至橫向平面。此時所產生的磁化稱為 M_{xy} ，而不再稱為 M_z ，因為它現在位於 xy 平面中。只要存在橫向磁化，它就會繞著 z 軸旋轉或進動；這種作用類似於電氣發電機，會在接收線圈中感應出頻率與拉莫爾頻率相同的交變電壓，即 MR 訊號。此訊號會由高靈敏度的接收器與電腦收集並處理，以產生 MR 影像。自旋系統的激發過程以圖形方式示於 ► 圖 5。

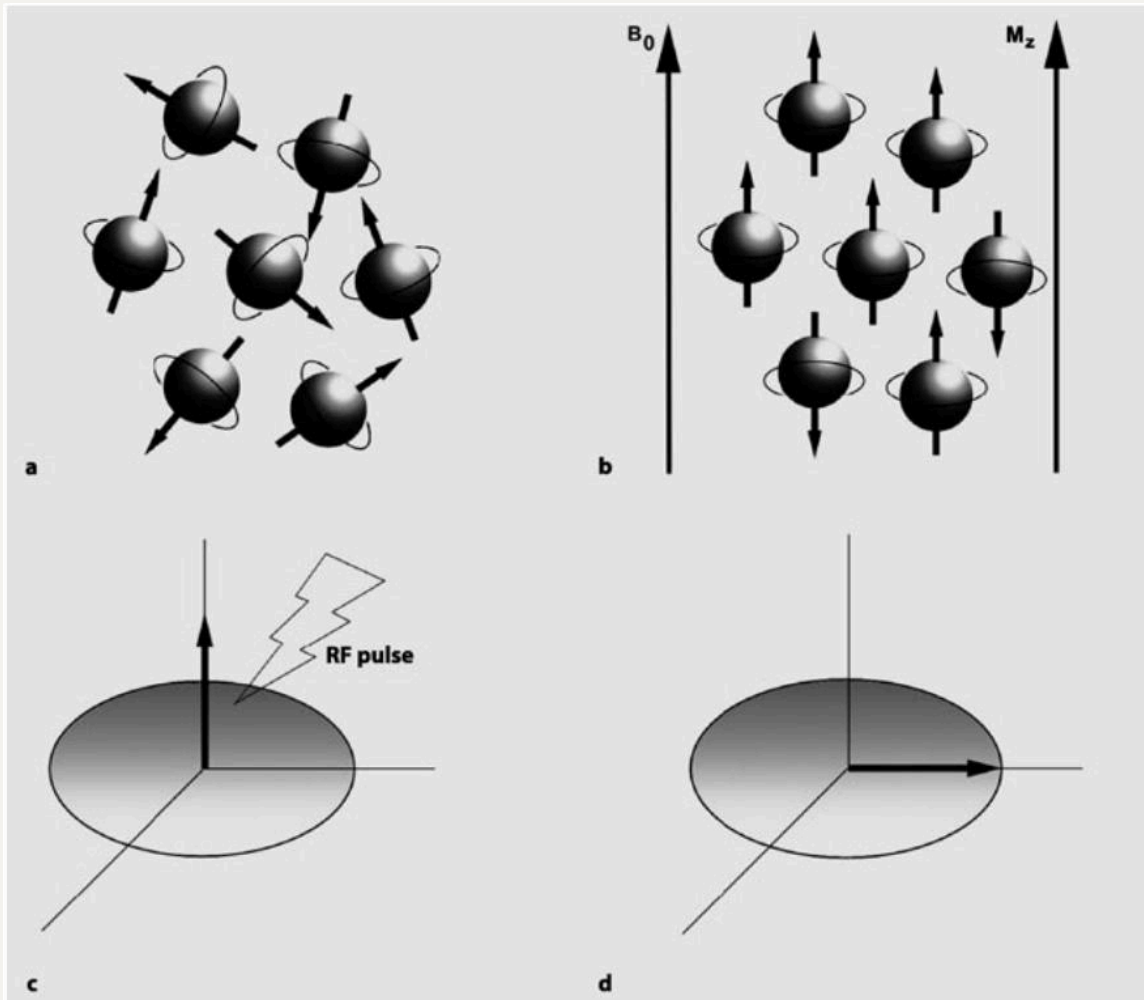


Fig. 5a–d. With no external magnetic field present, spins rotate about their axes in random direction (a). In the presence of a magnetic field, slightly more spins align parallel to the main magnetic field, B_0 , and thus produce longitudinal magnetization, M (b). An RF pulse (c) tips the magnetization vector by exactly 90° , causing the entire longitudinal magnetization to flip over and rotate into transverse magnetization, M_{xy} (d)

圖 5a–d。在不存在外部磁場時，自旋會以隨機方向繞其軸旋轉 (a)。在磁場存在時，會有稍多的自旋與主磁場 B_0 ，平行排列，因而產生縱向磁化強度 M (b)。RF 脈衝 (c) 會使磁化向量精確偏轉 90° ，導致整個縱向磁化翻轉並旋轉成橫向磁化強度 M_{xy} (d)

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